

Project Project– Draft Visualizations

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Install/Import packages

```
# install.packages("gamma")  
# install.packages("RColorBrewer")  
# install.packages("kableExtra")  
# BiocManager::install("org.Hs.eg.db")  
library(RColorBrewer)
```

```
## Warning: package 'RColorBrewer' was built under R version 4.5.2
```

```
library(gamma)
```

```
## Warning: package 'gamma' was built under R version 4.5.2
```

```
library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
## Warning: package 'BiocGenerics' was built under R version 4.5.2
```

```
## Loading required package: generics
```

```
##
```

```
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
```

```
##      setequal, union
```

```
##
```

```
## Attaching package: 'BiocGenerics'
```

```

## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##   order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##   rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##   unsplit, which.max, which.min

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##   findMatches

## The following objects are masked from 'package:base':
##
##   expand.grid, I, unname

## Loading required package: IRanges

##
## Attaching package: 'IRanges'

## The following object is masked from 'package:grDevices':
##
##   windows

## Loading required package: GenomicRanges

## Loading required package: GenomeInfoDb

## Loading required package: SummarizedExperiment

## Loading required package: MatrixGenerics

## Loading required package: matrixStats

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##   colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##   colCounts, colCummaxs, colCummins, colCumprods, colCumsums,

```

```
## colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
## colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
## colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
## colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
## colWeightedMeans, colWeightedMedians, colWeightedSds,
## colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
## rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
## rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
## rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
## rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
## rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
## rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
## rowWeightedSds, rowWeightedVars
```

```
## Loading required package: Biobase
```

```
## Welcome to Bioconductor
```

```
##
```

```
## Vignettes contain introductory material; view with
## 'browseVignettes()'. To cite Bioconductor, see
## 'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
##
```

```
## Attaching package: 'Biobase'
```

```
## The following object is masked from 'package:MatrixGenerics':
```

```
##
```

```
## rowMedians
```

```
## The following objects are masked from 'package:matrixStats':
```

```
##
```

```
## anyMissing, rowMedians
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:Biobase':
```

```
##
```

```
## combine
```

```
## The following object is masked from 'package:matrixStats':
```

```
##
```

```
## count
```

```
## The following objects are masked from 'package:GenomicRanges':
```

```
##
```

```
## intersect, setdiff, union
```

```
## The following object is masked from 'package:GenomeInfoDb':
##
## intersect

## The following objects are masked from 'package:IRanges':
##
## collapse, desc, intersect, setdiff, slice, union

## The following objects are masked from 'package:S4Vectors':
##
## first, intersect, rename, setdiff, setequal, union

## The following objects are masked from 'package:BiocGenerics':
##
## combine, intersect, setdiff, setequal, union

## The following object is masked from 'package:generics':
##
## explain

## The following object is masked from 'package:gamma':
##
## summarise

## The following objects are masked from 'package:stats':
##
## filter, lag

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2

##
## Attaching package: 'tidyr'

## The following object is masked from 'package:S4Vectors':
##
## expand
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(pheatmap)
```

```
## Warning: package 'pheatmap' was built under R version 4.5.2
```

```
library(RColorBrewer)
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.5.2
```

```
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.5.2
```

```
##
## Attaching package: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##   group_rows
```

```
library(org.Hs.eg.db)
```

```
## Loading required package: AnnotationDbi
```

```
##
## Attaching package: 'AnnotationDbi'
```

```
## The following object is masked from 'package:dplyr':
##
##   select
##
```

```
setwd("D:\\Git Repositories\\BINF-5503-Group-5-Poster-Project\\Poster Project\\Datasets")
```

The sequencing was performed on tumor cultures from 4 patients at 2 time points for a total of 8 count files, this data was previously cleaned using “PosterProject.py” but was pre-filtered just to ensure cleaning was correct, followed by DESeq differential expression analysis

```
cts <- read.csv("combinedData.tsv", sep = "\t", row.names="gene_id")
anot <- read.csv("dataset_annotation.csv")
# Performing DESeq analysis using Day
dds <- DESeqDataSetFromMatrix(cts, anot, ~ day)
```

```
## the design formula contains one or more numeric variables with integer values,
## specifying a model with increasing fold change for higher values.
## did you mean for this to be a factor? if so, first convert
## this variable to a factor using the factor() function
```

```
#Pre-filter
dim(dds)
```

```
## [1] 23006      8
```

```

smallestGroupSize <- 3
keep <- rowSums(counts(dds) >= 10) >= smallestGroupSize
dds <- dds[keep, ]
dim(dds)

```

```
## [1] 17240      8
```

```

# Run DESeq() and results()
dds_out <- DESeq(dds)

```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```

res <- results(dds_out)
# res
mcols(res)

```

```

## DataFrame with 6 rows and 2 columns
##               type      description
##      <character>    <character>
## baseMean      intermediate mean of normalized c..
## log2FoldChange results log2 fold change (ML..
## lfcSE          results  standard error: day
## stat           results  Wald statistic: day
## pvalue         results  Wald test p-value: day
## padj           results  BH adjusted p-values

```

```
summary(res)
```

```

##
## out of 17240 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 155, 0.9%
## LFC < 0 (down)    : 165, 0.96%
## outliers [1]      : 285, 1.7%
## low counts [2]     : 1003, 5.8%
## (mean count < 12)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results

```

```

resultsNames(dds)

## character(0)

#NA results are removed from the resulting differential expression analysis
res <- na.omit(res)

# How many of our genes are _significantly_ differentially expressed?
nrow(res[res$pvalue < 0.05, ])

## [1] 1395

# How many have large fold changes?
nrow(res[abs(res$log2FoldChange) > 1, ])

## [1] 8

# How many of our genes are differentially expressed (by our significance and logFC thresholds)?
nrow(res[res$padj < 0.05 & abs(res$log2FoldChange) > 1, ])

## [1] 8

# How many are upregulated?
nrow(res[res$padj < 0.05 & res$log2FoldChange > 1, ])

## [1] 2

# How many are down regulated?
nrow(res[res$padj < 0.05 & res$log2FoldChange < -1, ])

## [1] 6

# Performing Gene expression analysis using day + patient
dds <- DESeqDataSetFromMatrix(cts, anot, ~ day + patient)

## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors

## the design formula contains one or more numeric variables with integer values,
## specifying a model with increasing fold change for higher values.
## did you mean for this to be a factor? if so, first convert
## this variable to a factor using the factor() function

# Pre-filter
smallestGroupSize <- 3
keep <- rowSums(counts(dds) >= 10) >= smallestGroupSize
dds <- dds[keep, ]
dim(dds)

## [1] 17240      8

```

```
# Run DESeq()
dds_condition_type <- DESeq(dds)
```

```
## estimating size factors
## estimating dispersions
## gene-wise dispersion estimates
## mean-dispersion relationship
## final dispersion estimates
## fitting model and testing
```

```
res <- results(dds_condition_type)
summary(res)
```

```
##
## out of 17240 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 3537, 21%
## LFC < 0 (down)    : 3218, 19%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 4)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```
resultsNames(dds_condition_type)
```

```
## [1] "Intercept"          "day"                  "patient_A820_vs_A778"
## [4] "patient_A870_vs_A778" "patient_A899_vs_A778"
```

```
# Record Statistical difference based on Day and Patient
```

```
resCond1 <- results(dds_condition_type, name = "day")
resCond2 <- results(dds_condition_type, name = "patient_A820_vs_A778")
resCond3 <- results(dds_condition_type, name = "patient_A870_vs_A778")
resCond4 <- results(dds_condition_type, name = "patient_A899_vs_A778")

summary(resCond1)
```

```
##
## out of 17240 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 1959, 11%
## LFC < 0 (down)    : 2151, 12%
## outliers [1]      : 0, 0%
## low counts [2]    : 335, 1.9%
## (mean count < 9)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```



```
summary(resCond2)
```

```
##
## out of 17240 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 4320, 25%
## LFC < 0 (down)    : 4371, 25%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 4)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```
summary(resCond3)
```

```
##
## out of 17240 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 3698, 21%
## LFC < 0 (down)    : 3576, 21%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 4)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```
summary(resCond4)
```

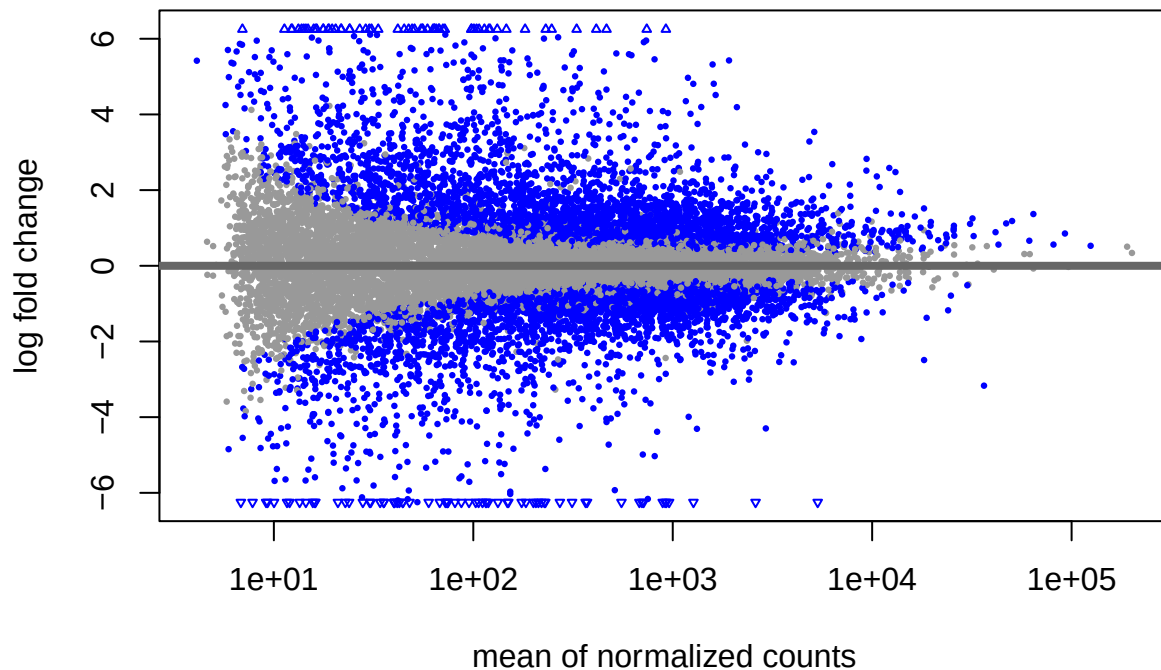
```
##
## out of 17240 with nonzero total read count
## adjusted p-value < 0.1
## LFC > 0 (up)      : 3537, 21%
## LFC < 0 (down)    : 3218, 19%
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 4)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```
# Determining Degrees of freedom
```

```
nrow(resCond1[!is.na(resCond1$padj) & resCond1$padj < 0.05 & abs(resCond1$log2FoldChange) > 1, ])
```

```
## [1] 7
```

```
plotMA(res)
```



```
#adding column to label genes as up/Down regulated or 'NA' if pvalue is insignificant
res_df <- data.frame(res) # fill in your results object
res_df$DEG <- "NA"
res_df$DEG[res_df$log2FoldChange > 1 & res_df$padj < 0.05] <- "Up-regulated"
res_df$DEG[res_df$log2FoldChange < -1 & res_df$padj < 0.05] <- "Down-regulated"
head(res_df)
```

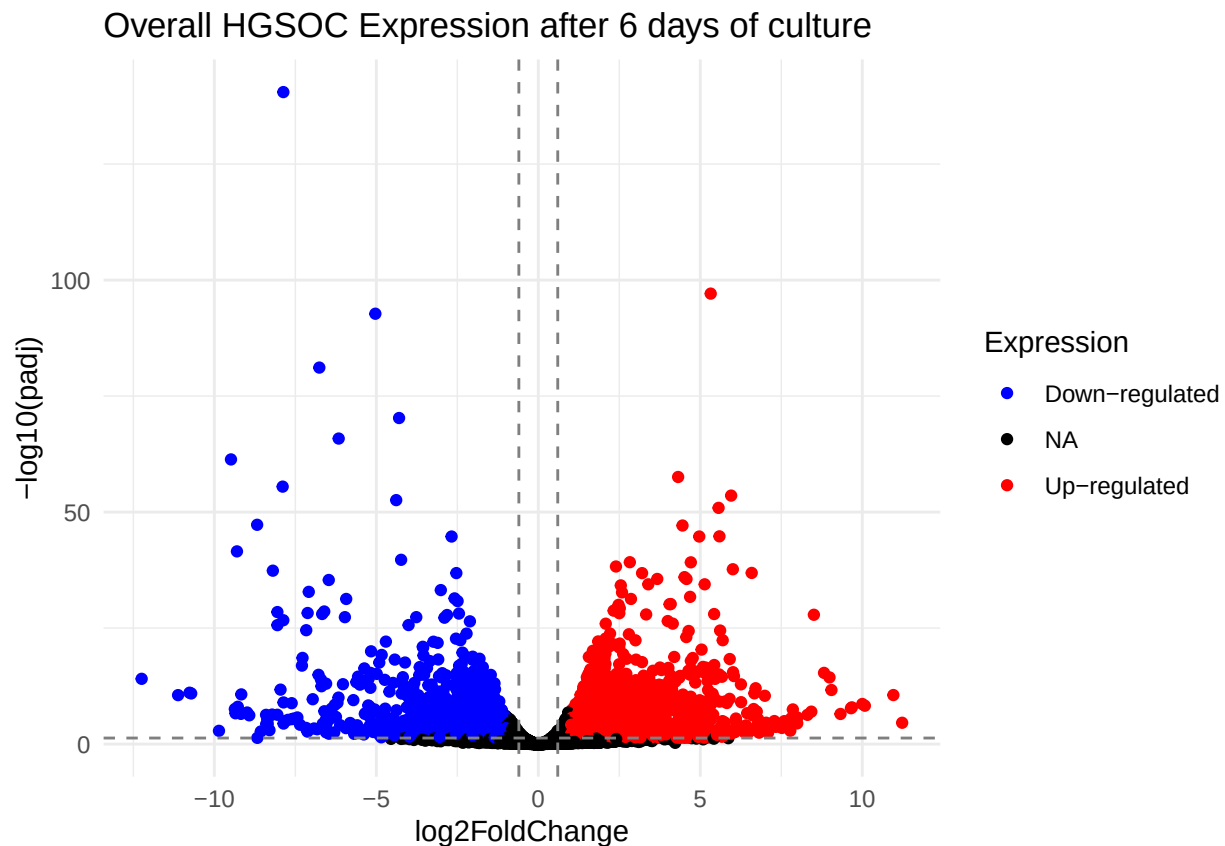
```
##          baseMean log2FoldChange    lfcSE      stat    pvalue
## ENSG00000000419.12 1102.23075   -0.14313119 0.2135938 -0.67010947 0.50278801
## ENSG00000000457.13  235.70203   -0.61055969 0.2789430 -2.18883293 0.02860899
## ENSG00000000460.16  241.03512   -0.44330526 0.4753095 -0.93266651 0.35099218
## ENSG00000000938.12   38.62925   -1.77323479 0.8685869 -2.04151677 0.04119949
## ENSG00000000971.15  828.37133   -0.82698194 1.3328842 -0.62044545 0.53496456
## ENSG00000001036.13 1085.47598   -0.01090209 0.3315437 -0.03288281 0.97376804
##          padj  DEG
## ENSG00000000419.12 0.64992617 NA
## ENSG00000000457.13 0.07820182 NA
## ENSG00000000460.16 0.51082554 NA
## ENSG00000000938.12 0.10400925 NA
## ENSG00000000971.15 0.67675035 NA
## ENSG00000001036.13 0.98542857 NA
```

```
table(res_df$DEG)
```

```
##
```

## Down-regulated	NA	Up-regulated
## 1638	13439	2163

```
ggplot(data = res_df, aes(x = log2FoldChange, y = -log10(padj), col = DEG)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Overall HGSOC Expression after 6 days of culture",
    colour = "Expression"
  ) +
  scale_color_manual(values = c("blue", "black", "red")) +
  geom_vline(xintercept = c(-0.6, 0.6), col = "grey50", linetype = 'dashed') +
  geom_hline(yintercept = -log10(0.05), col = "grey50", linetype = 'dashed')
```



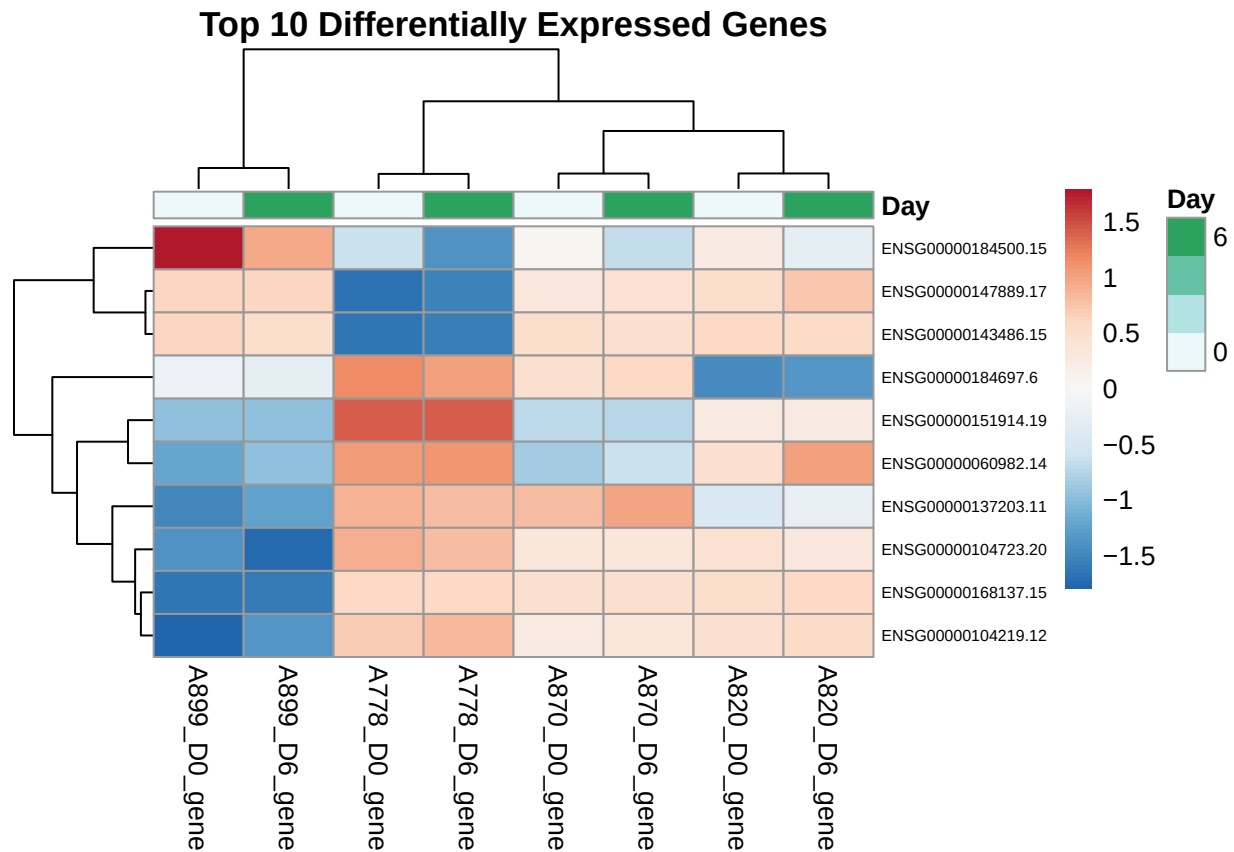
```
# vsd <- vst(dds, blind = FALSE) # variance stabilized transformation. Corrects for size factors or normalization
# sampleDists <- dist(t(assay(vsd))) # create a distance matrix between samples
# sampleDistMatrix <- as.matrix(sampleDists)
# rownames(sampleDistMatrix) <- paste(vsd$file)
# colnames(sampleDistMatrix) <- paste(vsd$file)
# pheatmap(sampleDistMatrix,
#           clustering_distance_rows = sampleDists,
#           clustering_distance_cols = sampleDists)
# Selecting the top 10 genes based on p-adjusted value
top_genes <- res_df %>%
  slice_min(padj, n=10)
print(top_genes)
```

##		baseMean	log2FoldChange	lfcSE	stat	pvalue
##	ENSG000000168137.15	2602.3780	-7.869665	0.3064067	-25.68373	1.776798e-145
##	ENSG000000147889.17	1584.6412	5.321277	0.2484782	21.41547	9.586974e-102
##	ENSG000000151914.19	812.0391	-5.029816	0.2403623	-20.92598	3.106024e-97
##	ENSG000000060982.14	5331.8400	-6.758877	0.3449183	-19.59559	1.686276e-85
##	ENSG000000184697.6	2929.3929	-4.297711	0.2352838	-18.26607	1.541497e-74
##	ENSG000000137203.11	750.1630	-6.161528	0.3482906	-17.69077	4.939693e-70
##	ENSG000000104723.20	929.2087	-9.486428	0.5551528	-17.08796	1.824498e-65
##	ENSG000000184500.15	626.9384	4.315578	0.2605415	16.56388	1.271248e-61
##	ENSG000000104219.12	714.0146	-7.890844	0.4850995	-16.26644	1.707919e-59
##	ENSG000000143486.15	735.5668	5.952506	0.3723860	15.98477	1.631560e-57
##		padj			DEG	
##	ENSG000000168137.15	3.063199e-141			Down-regulated	
##	ENSG000000147889.17	8.263972e-98			Up-regulated	
##	ENSG000000151914.19	1.784928e-93			Down-regulated	
##	ENSG000000060982.14	7.267850e-82			Down-regulated	
##	ENSG000000184697.6	5.315083e-71			Down-regulated	
##	ENSG000000137203.11	1.419338e-66			Down-regulated	
##	ENSG000000104723.20	4.493478e-62			Down-regulated	
##	ENSG000000184500.15	2.739540e-58			Up-regulated	
##	ENSG000000104219.12	3.271613e-56			Down-regulated	
##	ENSG000000143486.15	2.812810e-54			Up-regulated	

```

norm_counts <- counts(dds, normalized = FALSE)
mat <- norm_counts[rownames(top_genes), ]
mat <- log2(mat + 1)
mat_scaled <- t(scale(t(mat)))
annotation_col <- data.frame(
  Day = colData(dds)$day
)
rownames(annotation_col) <- colnames(mat_scaled)
pheatmap(mat_scaled,
  color = colorRampPalette(rev(brewer.pal(n = 7, name = "RdBu")))(100),
  annotation_col = annotation_col,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_rownames = TRUE,
  show_colnames = TRUE,
  fontsize_row = 6,
  main = "Top 10 Differentially Expressed Genes")

```



```
# for table formatting
gene_ids <- sub("\\..*", "", rownames(top_genes)) # remove ensemble id version number
ensembl_ids <- AnnotationDbi::select(org.Hs.eg.db,
                                     keys = gene_ids,
                                     column = c("SYMBOL", "GENENAME"),
                                     keytype = "ENSEMBL")
```

```
## 'select()' returned 1:1 mapping between keys and columns
```

```
ensembl_ids %>%
  mutate(DEG = top_genes$DEG) %>%
  kbl() %>%
  kable_classic(full_width = F, html_font = "Cambria")
```

Performing PCA analysis

```
#' Changing top_genes to include 1000 top diff genes used in the DEseq Differential
#' expression analysis
top_genes <- res_df %>%
  slice_min(padj, n=1000)
mat <- norm_counts[rownames(top_genes), ]
```

ENSEMBL	SYMBOL	GENENAME	DEG
ENSG00000168137	SETD5	SET domain containing 5	Down-regulated
ENSG00000147889	CDKN2A	cyclin dependent kinase inhibitor 2A	Up-regulated
ENSG00000151914	DST	dystonin	Down-regulated
ENSG00000060982	BCAT1	branched chain amino acid transaminase 1	Down-regulated
ENSG00000184697	CLDN6	claudin 6	Down-regulated
ENSG00000137203	TFAP2A	transcription factor AP-2 alpha	Down-regulated
ENSG00000104723	TUSC3	tumor suppressor candidate 3	Down-regulated
ENSG00000184500	PROS1	protein S	Up-regulated
ENSG00000104219	ZDHHC2	zinc finger DHHC-type palmitoyltransferase 2	Down-regulated
ENSG00000143486	EIF2D	eukaryotic translation initiation factor 2D	Up-regulated

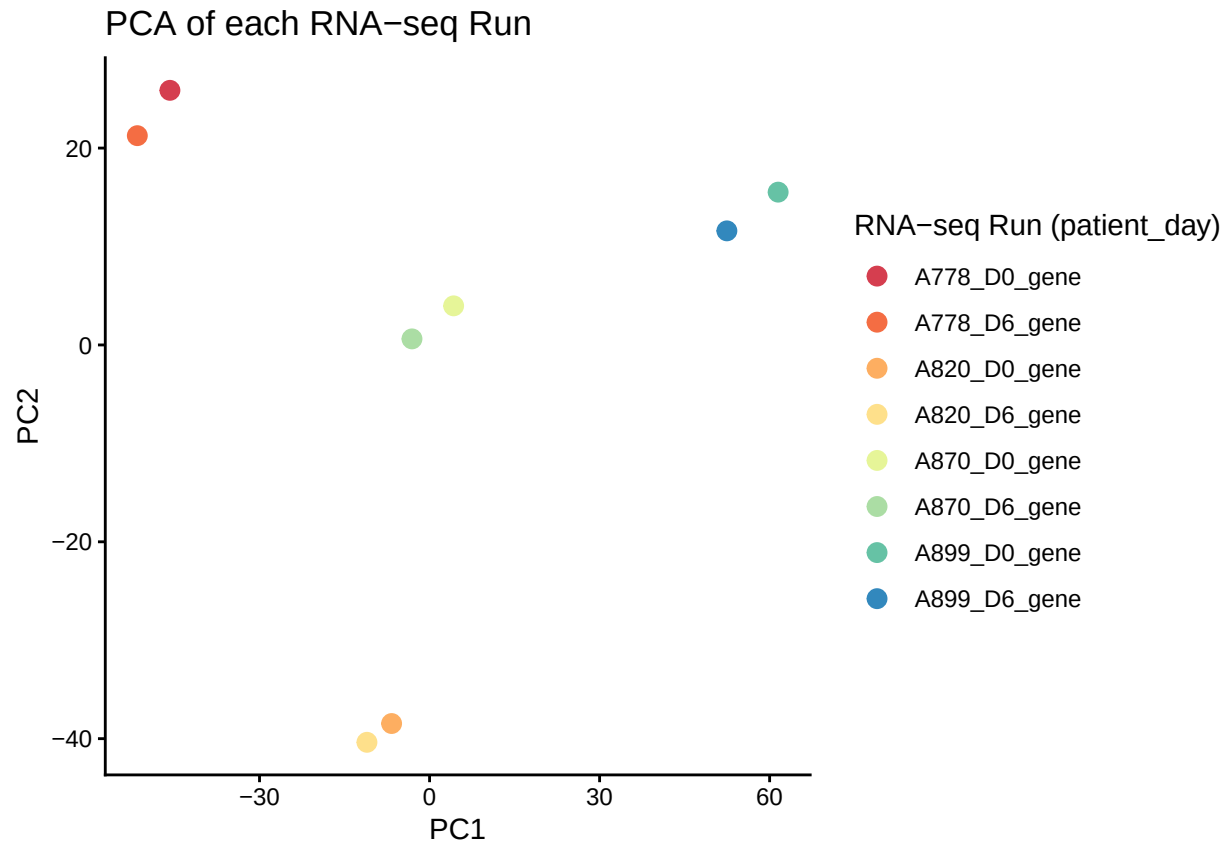
```

mat <- log2(mat + 1)
mat_scaled <- t(scale(t(mat)))
annotation_col <- data.frame(
  Day = colData(dds)$day
)
rownames(annotation_col) <- colnames(mat_scaled)

# expression_matrix: genes x samples seq (RNA Seq File)
pca <- prcomp(t(mat), scale. = FALSE)
pca_df <- data.frame(
  PC1 = pca$x[,1],
  PC2 = pca$x[,2],
  Day = colData(dds)$day,
  Patient = colData(dds)$patient
)

ggplot(pca_df, aes(x = PC1, y = PC2, colour = row.names(pca_df))) +
  geom_point(size = 3) +
  theme_classic() +
  labs(
    x = paste0("PC1"),
    y = paste0("PC2"),
    title = "PCA of each RNA-seq Run",
    colour = "RNA-seq Run (patient_day)"
  ) +
  scale_color_brewer(palette="Spectral")

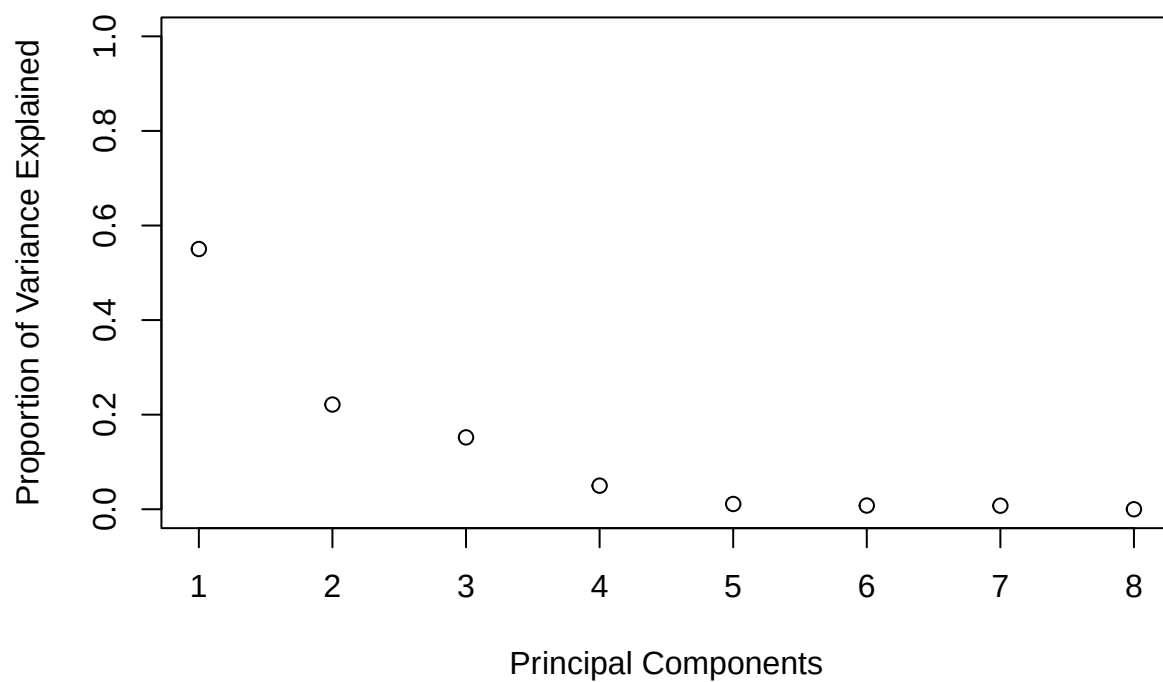
```



```
# expression_matrix: splitting matrix based on patient and day (RNA Seq File)
pca_df2 <- pca_df %>%
  mutate(
    Day = as.character(Day),
    Patient = as.character(Patient)
  ) %>%
  pivot_longer(cols = c(Day, Patient),
               names_to = "ColorBy",
               values_to = "Group")

# plot PC variance in a scree plot
variance_pca = pca$sdev^2
# proportion variance
variance_pca / sum(variance_pca) -> prop_pca

plot(prop_pca,
     xlab = "Principal Components",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1))
```



```
ggplot(pca_df2, aes(x = PC1, y = PC2, colour = Group)) +
  geom_point(size = 3) +
  theme_classic()+
  facet_wrap(~ ColorBy, scales = "free") +
  labs(
    title = "PCA of RNA-seq Samples",
    x = paste0("PC1 (", round(100 * summary(pca)$importance[2,1], 1), "%)"),
    y = paste0("PC2 (", round(100 * summary(pca)$importance[2,2], 1), "%)"),
    colour = "Day Patient"
  )+
  scale_color_brewer(palette="Set1")
```